Predicting Airbnb Occupancy and Pricing Using Listing and Neighborhood Demographic Data

Machine Learning Project Proposal

Authors: Roman Shrestha and Tanish Pradhan Wong Ah Sui

Introduction

The short term rental market, disrupted by Airbnb in the early 2010s has grown into a major economic force in regions across the world. Airbnb today comprises of millions of listings globally and generate substantial income for hosts while also influencing housing markets. However the platform's algorithmic systems driven by machine learning, that determine which listings receive prominence in search results play an important role, which often lacks transparency, in shaping the success of listings. In Professor Stapleton's *Data and its Discontents* class, we learned, by examining prior research, that algorithmic recommendation systems on digital platforms have the capability and tend to reinforce existing disparities and unequal outcomes.

There is a growing body of research around this topic that we aim to build on top of. Research by Edelman, Luca, and Svirsky (2017), through a field experiment on Airbnb, found that guest applications from individuals with distinctively African American names were 16% less likely to be accepted relative to identical guests with distinctively white names, with discrimination occurring across landlords of all sizes and suggesting that Airbnb's platform design actively facilitates racial bias. Building on this, a large scale study by Ke et al. (2022), examining over 834,000 listings, 513,000 hosts, and 13 million reviews across 97 cities, found that Black hosts receive on average 22% less revenue than comparable white hosts, with machine learning classification of host profile pictures used to estimate the racial revenue gap. Together, these studies suggest that Airbnb activity is not evenly distributed across urban neighborhoods and may reflect and reinforce existing patterns of economic segregation.

We will use machine learning models to examine how listing characteristics and neighborhood level demographic factors together predict Airbnb occupancy rates and pricing. By incorporating US Census demographic data, including measures of income, poverty, race, and housing, along with platform listing data, we aim to gain insight into what factors are most associated with listing success and whether and to what extent those factors amplify demographic disparities.

Motivation

As international students from the Global South, housing costs and real estate inequality have always been tangible real issues that have surrounded us. Growing up in poverty stricken regions in Nepal and Northeast India, from our own experience, we have seen how gentrification displaces long standing communities and how the benefits of economic growth are divided unevenly along racial and class lines. After coming to study in the US, we found many of the same dynamics and inequality but at a higher

technological level increasingly mediated by digital platforms that operate with little transparency about how they shape economic outcomes.

That is why we hope to better understand how algorithmic promotion and market forces interact with neighborhood level demographics on a relevant major platform like Airbnb.

Research Questions

1. Can machine learning models accurately predict which Airbnb listings result in higher occupancy rates, and which listing factors are most predictive?
2. What listing and host characteristics are most strongly associated with Airbnb pricing in the US?
3. Do neighborhood level demographics including household income, poverty rate and racial composition significantly predict occupancy rates and prices, independent of or in addition to listing level attributes?

Data Description

The primary source of our data is the Inside Airbnb dataset, an open-source data which consists of listing, occupancy, review and neighborhood information from Airbnb listings in major cities of the world from 2025. We combined this data with demographic information of the neighbourhood where the airbnb listings are based, which comes from the American Community Survey (ACS) 2023. In the future, we also plan to incorporate the FBI Crime Data Explorer dataset as we believe it will be a significant predictor for listing outcomes across neighborhoods.

After cleaning and merging the data, our final dataset has 280,286 observations and 61 variables. Each observation in the dataset is a unique airbnb listing present in 2025 from one of the following US cities: Los Angeles, New York City, Hawaii, Clark County, Broward, San Diego, Austin, Nashville, Chicago, San Francisco, New Orleans, Seattle, Santa Clara, Washington D.C, Dakkas, Rhode Island, Twin Cities, Denver, Portland, Boston, San Mateo, Columbus, Asheville, Oakland, Fort Worth, Newark, Jersey City, Santa Cruz, Cambridge, Rochester, Bozeman, Albany, Salem, and Pacific Grove. LA, New York City and Hawaii are the top three areas with the highest share of US based airbnb listings (16%, 12.9%, and 11.9% respectively). Detailed share of each US city is listed in the appendix section.

The data dictionary table below lists the variables we have in the dataset and describes them briefly. Broadly, the variables can be categorised into 6 key categories: dependent variables (e.g. price, occupancy), listing features (e.g. property type, bedrooms), host features (e.g. host_response_time, host_acceptance_rate), location (e.g. city, neighbourhood_cleansed), and census demographics (e.g. median_household_income, median_gross_rent).

Variable	Description	Type
price	Nightly listing price in USD	Integer
estimated_occupancy_l365d	Estimated nights booked in the past 365 days	Float
estimated_revenue_l365d	Estimated revenue earned in the past 365 days (USD)	Float
review_scores_rating	Overall guest rating, scale 0–5	Float
reviews_per_month	Average reviews received per month	Float
room_type	Airbnb room type (entire home/apt, private room, hotel room)	Categorical
accommodates	Max number of guests the listing can accommodate	Integer
bedrooms	Number of bedrooms	Float
beds	Number of beds	Float
bathrooms	Number of bathrooms	Float
minimum_nights	Minimum nights required to book	Integer
maximum_nights	Maximum nights allowed	Integer
instant_bookable	Whether listing can be booked without host approval	Boolean
availability_365	Days available in the next 365 days	Float
availability_30	Days available in the next 30 days	Float
availability_60	Days available in the next 60 days	Float
availability_90	Days available in the next 90 days	Float
minimum_minimum_nights	Minimum value of minimum nights across the booking window	Float
has_wifi	Whether the listing offers wifi	Boolean
has_parking	Whether free parking is available	Boolean
has_pool	Whether a pool is available	Boolean
has_gym	Whether a gym is available	Boolean
host_id	Unique identifier for the host	Integer
host_is_superhost	Whether the host has superhost status	Boolean
host_response_time	Ordinal: 1=within an hour, 2=few hours, 3=day, 4=few days, 0=unknown	Ordinal
host_response_rate	Share of inquiries the host responded to, 0–1	Float
host_acceptance_rate	Share of booking requests the host accepted, 0–1	Float
host_identity_verified	Whether host identity has been verified by Airbnb	Boolean
host_has_profile_pic	Whether the host has a profile picture	Boolean

hosts_time_as_host_years	Number of years the user has been a host	Float
number_of_reviews	Total reviews the listing has received	Float
review_scores_cleanliness	Guest rating for cleanliness, scale 0–5	Float
review_scores_location	Guest rating for location, scale 0–5	Float
review_scores_value	Guest rating for value, scale 0–5	Float
review_scores_communication	Guest rating for host communication, scale 0–5	Float
city	City name derived from source file (e.g. albany_NY)	Categorical
latitude	Latitude coordinate of the listing	Float
longitude	Longitude coordinate of the listing	Float
neighbourhood_cleansed	Standardized neighborhood name	Categorical
full_tract_fips	11-digit Census tract FIPS code	String
state_fips	2-digit state FIPS code	String
county_fips	3-digit county FIPS code	String
tract_fips	6-digit tract FIPS code	String
listing_url	Airbnb URL for the listing	String
name	Listing title	String
description	Full listing description	String
median_household_income	Median household income in the census tract (USD)	Float
median_gross_rent	Median gross rent in the census tract (USD)	Float
median_home_value	Median home value in the census tract (USD)	Float
total_population	Total population of the census tract	Float
median_age	Median age of residents in the census tract	Float
poverty_rate	Share of tract population below poverty line, 0–1	Float
pct_renter_occupied	Share of housing units that are renter-occupied, 0–1	Float
pct_white	Share of tract population identifying as white non-Hispanic, 0–1	Float
pct_black	Share of tract population identifying as Black, 0–1	Float
pct_asian	Share of tract population identifying as Asian, 0–1	Float
pct_hispanic	Share of tract population identifying as Hispanic/Latino, 0–1	Float
pct_college_educated	Share of adults 25+ with a bachelor's degree or higher, 0–1	Float
unemployment_rate	Share of civilian labor force that is unemployed, 0–1	Float

Table 1: Data Dictionary

The summary statistics about the numerical variables is presented in the table below.

	mean	std	min	50%	max
price	221.52	182.8	0	161	767.95
estimated_occupancy_l365d	73.12	88.9	0	30	255
estimated_revenue_l365d	14553.83	16823.27	0	8160	57626.1
review_scores_rating	4.78	0.39	0	4.89	5
reviews_per_month	1.5	1.86	0.01	0.91	120.37
accommodates	4.4	3.04	1	4	16
bedrooms	1.79	1.34	0	1	57
beds	2.32	1.94	0	2	153
maximum_nights	264.83	145.93	1	365	365
availability_365	213.66	128.46	0	249	365
availability_30	14.03	11.4	0	14	30
availability_60	32.34	22.13	0	36	60
availability_90	52.31	32.75	0	61	90
minimum_minimum_nights	13.71	26.77	0	2	1125
host_response_time	1.07	0.78	0	1	4
calculated_host_listings_count	36.55	112.26	1	3	1189
hosts_time_as_host_years	6.09	3.72	0	6	14
number_of_reviews	49.38	97.08	0	11	3920
review_scores_cleanliness	4.76	0.4	0	4.88	5
review_scores_location	4.8	0.35	0	4.9	5
review_scores_value	4.7	0.42	0	4.8	5
review_scores_communication	4.86	0.35	0	4.96	5
median_household_income	101329.93	44278.12	2499	94194	250001
median_gross_rent	2019.76	644.52	239	1939	3501
median_home_value	835703.93	471468.76	9999	721800	2000001
total_population	3932.92	1815.83	6	3716	24659
poverty_rate	0.13	0.09	0	0.11	1

pct_renter_occupied	0.55	0.24	0	0.56	1
pct_white	0.48	0.24	0	0.5	1
pct_black	0.12	0.18	0	0.04	1
pct_asian	0.13	0.14	0	0.08	1
pct_hispanic	0.2	0.18	0	0.14	1
pct_college_educated	0.51	0.21	0	0.52	1
unemployment_rate	0.06	0.04	0	0.05	1
bathrooms	1.55	0.99	0	1	50

Table 2: Summary Statistics of Numerical Variables

The categorical variables also give us a good understanding of the airbnb data. We find that there are four major property categories that get listed: entire home, private room, hotel, and share rooms, with entire homes constituting 75.9% of the total listings, followed by 22.5% private rooms, 1.1% hotels and 0.5% share rooms. Only 35.4% of the listings are instantly bookable, while the remaining require approval from the host. 98.6% of these listings have wifi, 54.3% have free parking, 29.9% have a pool, and 17.6% have a gym.

The figure below shows the distribution of some of the numerical variables. We see that price is right skewed with most listings concentrated between $50 to $300 a night. The estimated revenue and estimated occupancy also follow the similar right-skewed pattern. We see a small cluster of high occupancy listings, which are likely professionally managed listings. The review scores seem to be strong left skewed and most reviews lie in the 4 to 5 range. The median household income is roughly normally distributed and centered around $90,000. Finally we see that the poverty rate is right skewed with most tracts below 0.2.

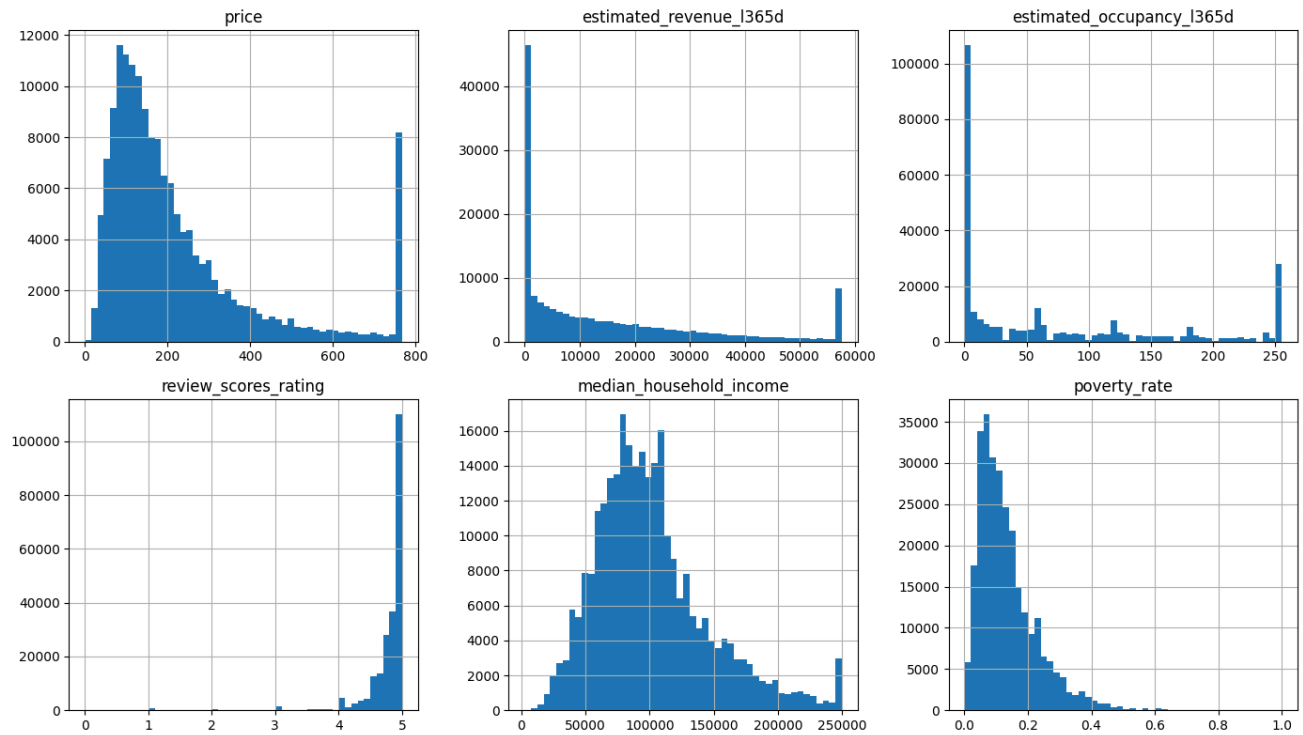

Figure 1: Distribution of key numerical variables

Data Cleaning and Preparation

We started with the data collection. Our primary source of data was insideairbnb.com. We downloaded U.S cities' listings from 2025 from the inside airbnb website and concatenated them into a single dataframe. To complement the data with demographic information, we first geo-coded the airbnb listings using their latitude and longitude information and extracted their census tracts. We then used these census tracks to make API calls to the census website to get the 2023 ACS5 data for the variables we plan to use in our analysis (for example, earnings data, rent data, etc). Once this data was acquired, we merged it with the census data. This resulted in airbnb listings that also contained demographic information about the neighborhoods they were in.

The next step was to clean the data. We first filtered the data to only keep the columns we planned to use in our analysis. We dropped 41 columns, the majority of which were id and text data that were irrelevant for our analysis, which resulted in 61 columns in our dataset. Then we converted the string boolean indicators (such as T, F) into boolean values. Next, we detected multiple outliers (such as columns like median income having negative values, listing price being extraordinarily high, etc), and resolved it by clipping the numbers at 0 (to keep them positive) and at the 95 percentile. We also extracted numerical information from columns such as bathroom which had both text and numbers in it. The amenities column was filtered to exclude for integer outliers (less than 0.1% of total observations), and dummy variables were created that indicated whether the listings had basic amenities such as wifi, parking, pool and gym. We then cleaned the host variables, which included steps like converting the text data to ordinal values,

and replacing the null values with 0 (for plotting purposes). We were unable to find demographic information for a few airbnbs (less than 0.1%), so we decided to drop them.

A lot of our time during this proposal was spent on matching the census tracts to listings and acquiring the census data, which was very slow due to rate limits from the census API. So, we plan to make more improvements to the data quality in future. One such improvement would be to deal with missing values in the price column. Approximately 43% of the price data is missing in the dataset. However, since price could be a good predictor variable for our analysis, we do not want to drop it entirely. We have two ideas we plan to explore: one is to see if certain cities have price variable missing more than others, and if that is the case, drop those cities from analysis; another is to impute neighbourhood or city median prices to the missing values.

Methods

After performing exploratory data analysis (EDA) including distribution plots, correlation matrices and geographic visualizations, which will give us a better sense of the data we are dealing with, we will create predictive models, which is the key part of our study.

For our primary predictive models, we were thinking of using the estimated occupancy rate and nightly price (log-transformed to avoid right skewed data) as outcome variables. We will train models using an 80/20 train-test split with cross-validation on the training set. Our modeling approach will proceed in stages:

First, we will fit linear and logistic regression baselines (OLS for price, logistic for a binary high/low occupancy outcome) to establish benchmarks and evaluate coefficient signs for the demographic variables. This also allows a first test of whether neighborhood demographics predict outcomes in addition to listing attributes.

Second, we will fit Random Forest models for both outcomes. We will tune the number of trees, maximum depth, and minimum samples per leaf using grid search with 5-fold cross-validation. Feature importances from the Random Forest will help identify which variables, including demographic variables, are most useful in predicting the outcomes.

Third, we will fit Gradient Boosted Trees (XGBoost or LightGBM) to capture non-linearity between listing and neighborhood features. We will be tuning hyperparameters like learning rate, max depth, number of estimators using Bayesian optimization or grid search.

To understand the contribution of demographic variables, we will compare model and feature importance by training models both with and without the demographic variables. We will also use SHAP values to gain insight into marginal feature contributions across models, which is essential for understanding how demographic features influence predictions.

Finally, if demographic features, particularly racial composition and income, remain significant predictors of occupancy or price even after controlling for listing quality, we will conduct additional analyses to see

if these patterns vary across regions and whether they are concentrated in specific listing or host types. This will let us understand more about where and how and what shape demographic disparities take on the platform.

Research Expectations

We expect that listing level features such as room type, listing rating, host rating and superhost status will be among the strongest predictors of both price and occupancy. We anticipate that gradient boosted models will outperform linear baselines due to the non-linearity of some factors. More importantly, we expect to find that neighborhood demographic characteristics particularly median household income, racial composition, and renter occupied housing share will be associated with listing performance even after controlling for listing attributes. If supported, these findings would suggest that Airbnb's marketplace outcomes reflect and may reinforce existing patterns of racial and socioeconomic inequality across neighborhoods in the US, with implications not only for individual hosts but for broader questions of economic opportunity, neighborhood investment, and the fairness of algorithmic platforms in housing markets.

Citations

Edelman, B., Luca, M., & Svirsky, D. (2017). Racial discrimination in the sharing economy: Evidence from a field experiment. American Economic Journal: Applied Economics, 9(2), 1–22. https://doi.org/10.1257/app.20160213

Ke, Q., et al. (2022). How sharing is the "sharing economy"? Evidence from 97 Airbnb markets. PLOS ONE. https://doi.org/10.1371/journal.pone.0266998

Appendix

	missing_n	missing_pct
hosts_time_as_host_years	189013	67.34
price	121034	43.12
estimated_revenue_l365d	121034	43.12
review_scores_communication	58176	20.73
review_scores_value	58197	20.73
review_scores_location	58200	20.73
review_scores_rating	58155	20.72
reviews_per_month	58167	20.72
review_scores_cleanliness	58170	20.72
beds	53248	18.97
host_response_time	47661	16.98
host_response_rate	47659	16.98
host_acceptance_rate	40417	14.40
bedrooms	14100	5.02
description	5775	2.06
host_is_superhost	5750	2.05
host_identity_verified	1528	0.54
host_has_profile_pic	1518	0.54
bathrooms_text	1167	0.42
neighbourhood_cleansed	1084	0.39
pct_renter_occupied	426	0.15
poverty_rate	402	0.14
unemployment_rate	353	0.13
pct_college_educated	328	0.12
pct_white	320	0.11
pct_black	320	0.11
pct_asian	320	0.11
pct_hispanic	320	0.11
minimum_minimum_nights	173	0.06
state_fips	30	0.01
median_age	30	0.01
total_population	30	0.01
median_home_value	30	0.01
median_gross_rent	30	0.01
median_household_income	30	0.01

tract_fips	30	0.01
county_fips	30	0.01
availability_365	16	0.01
full_tract_fips	30	0.01
availability_30	16	0.01
availability_60	16	0.01
availability_90	16	0.01
instant_bookable	15	0.01
maximum_nights	15	0.01
minimum_nights	15	0.01
calculated_host_listings_count	15	0.01
amenities	15	0.01
estimated_occupancy_l365d	15	0.01
accommodates	15	0.01
room_type	15	0.01
property_type	15	0.01
city	15	0.01
latitude	15	0.01
longitude	15	0.01
number_of_reviews	16	0.01
listing_url	3	0.00
name	3	0.00

Table 3: Percentage of missing data in variables

city	count	pct
la_CA	45423	16.2
nyc_NY	36080	12.9
hawaii_hawaii	33450	11.9
clark_NV	17624	6.3
broward_FL	16822	6.0
sandiego_CA	13160	4.7
austin_TX	10533	3.8
nashville_TN	9443	3.4
chicago_IL	8659	3.1
sf_CA	7535	2.7
neworleans_LA	7440	2.7
seatle_WA	6996	2.5
santaclara_CA	6940	2.5
washington_DC	6374	2.3
dallas_TX	5820	2.1
rhodeisland_RI	5762	2.1
twincities_MN	5318	1.9
denver_CO	4909	1.8
portland_OR	4455	1.6
boston_MA	4399	1.6
sanmateo_CA	3515	1.3
columbus_OH	2877	1.0
asheville_NC	2852	1.0
oakland_CA	2250	0.8
fortworth_TX	2081	0.7
newark_NJ	1868	0.7
jerserycity_NJ	1829	0.7
santacruz_CA	1760	0.6
cambridge_MA	1358	0.5
rochester_NY	1032	0.4
bozeman_MT	600	0.2
albany_NY	478	0.2
salem_OR	351	0.1
pacificgrove_CA	293	0.1

Table 4: Listings by city